**BRAINWARE UNIVERSITY**
**SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES**
**DEPARTMENT OF COMPUTATIONAL SCIENCES**
**BACHELOR OF COMPUTER APPLICATIONS /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024**
**SEMESTER –V**

**Course Code: BCA50112**
**Course Name: Software Engineering**
**Weekly Contact Hours: 3L+1T**
**Credits: 4**
**Total Allotted Hours: 45L+15T**

**Course Objective:** The aim of this course is **t**o know and apply basic principles of software engineering to different several applications and to apply the expertise to design, development and testing.

**Pre-requisite(s):** Knowledge of Software and system development principles.

**Course Outcome (COs):** After the completion of the course, students would be able to:

**CO1:** Define the principles of software engineering concept and understand the software development life cycle.
**CO2:** Explain the significance of different software models and identify their working principles.
**CO3:** Discuss about knowledge of cost model and its application.
**CO4:** Compare different testing methodologies and their application scopes.
**CO5:** Illustrate and criticize the software design strategies and maintenance steps.

**Module 1: Introduction to Software Engineering [3H]**

Definition of software, types of software, characteristics of software, attributes of good software, define software engineering, software engineering costs, key challenges faced by software engineers, systems engineering concept.

**Module 2: Software Development Process Models [10H]**

Software process, software process model, the waterfall model, evolutionary development, component-based software engineering (CBSE), process iteration, incremental delivery, spiral development, rapid software development, agile methods, extreme programming, rapid application development (RAD), software prototyping, computer aided software engineering (CASE), overview of CASE approach, classification of CASE tools.

**Module 3: Requirement Analysis and Specifications [10H]**

Software requirement analysis and specification, system and software requirements, types of software requirements ,functional and non-functional requirements, domain requirements, user requirements, elicitation and analysis of requirements ,overview of techniques, viewpoints interviewing scenarios, use-cases, process modeling with physical and logical DFDs, entity relationship diagram, data dictionary, requirement validation, requirement specification, software requirement specification (SRS), structure and contents SRS format, feasibility.

**Module 4: Software Design and Development Tools** **[10H]**

Software design concepts, abstraction, architecture, patterns, modularity, cohesion, coupling, information hiding, functional independence, refinement, design of input and control design of user interface design, elements of good design, design issues, features of modern GUI, menus, scroll bars, windows, buttons, icons, panels, error messages etc. coding, programming languages and development tools, selecting languages and tools, good programming practices, coding standards.

**Module 5: Software Testing and Project Management** **[12H]**

Software testing and quality assurance, verification and validation, different types of testing, design of test cases, quality management activities, product and process, quality standards ISO9000, capability maturity model (CMM),managing software projects, need for the proper management of software, project management activities, project planning, software size estimation and cost estimation, function point analysis ,LOC estimation , COCOMOII model, project scheduling task set for software project, defining a task ,network scheduling, earned value analysis, risk management systems .The RMMM managing people.

**Course Outcomes (COs) and Modules Mapping:**

| Sl. No. | Course outcome (COs) | Mapped Module(s) | Learning levels (as per Bloom's Taxonomy from K1 to K6) |
|---|---|---|---|
| **1.** | CO1 | M1 | K1, K2 |
| **2.** | CO2 | M2 | K2, K3 |
| **3.** | CO3 | M3 | K3, K6 |
| **4.** | CO4 | M4 | K3, K4 |
| **5.** | CO5 | M5 | K2,K6 |

**Text books:**

1. Fundamentals of Software Engineering, Rajib Mall, PHI Learning Pvt. Ltd, 5th Edition, 2018.

**Reference books:**

1. Software engineering, I. Sommerville, Addison-Wesley Longman,10th Edition,2016.
2. Software engineering: A precise approach, Pankaj Jalote, Wiley-India, 2nd Edition, 2010.
3. Software Engineering: A Practitioner's Approach, Pressman, McGraw Hill Publication, 9th Edition, 2020.